Supporting Information for Correlation-consistent Gaussian basis sets for copper solids from material-constrained atomic optimization

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CONTENTS

I. GEOMETRY AND BASIS-SET DATA

Geometry files for the CO/Cu(111) adsorption systems and data files for all Gaussian basis sets used in this work are available in the following GitHub repository:

https://github.com/hongzhouye/supporting_data/tree/main/2026/MCAO_Cu

The basis-set data include

- the MCAO-cc-pVXZ sets optimized in this work;
- the truncated def2-SVP and def2-TZVP sets generated following the protocol of ref 1; and
- the TZV2P-MOLOPT basis sets from ref 2.

For each orbital basis set, the repository also provides a corresponding auxiliary basis set optimized for density fitting using the protocol described in section IV.

II. COMPUTATIONAL DETAILS

This section summarizes the computational settings used throughout this work. Details for basis-set generation, auxiliary basis-set optimization, bulk calculations, and surface calculations are given in sections III to VI.

A. PySCF

All Gaussian-based calculations are performed using the PySCF code.[3,4] The RPA total energy is the sum of three terms, all evaluated using PBE orbitals,

$$E^{\mathrm{RPA}} = E^{\mathrm{HF}} + \Delta E^{\text{ACFD-X}} + E_{\mathrm{corr}}^{\mathrm{RPA}} \tag{1}$$

where E^{HF} is the HF energy, $\Delta E^{\text{ACFD-X}}$ is the difference between the adiabatic-connection fluctuation-dissipation (ACFD) exchange energy and the HF exchange energy discussed in ref 5, and $E_{\mathrm{corr}}^{\mathrm{RPA}}$ is the RPA correlation energy evaluated within the ACFD approach using 40 quadrature points for the imaginary-frequency integration. For the molecular dimer RPA calculations, the PBE reference is calculated at zero temperature without smearing, and the $\Delta E^{\text{ACFD-X}}$ term vanishes. For periodic bulk Cu and Cu surface RPA calculations, the PBE reference is calculated with the Fermi-Dirac smearing with a width of 0.01 eV, so the $\Delta E^{\text{ACFD-X}}$ term is nonzero whenever fractional occupations are present. For all nuclear potentials, the correlation energy is calculated by freezing the [Ar] core for Cu and the [He] core for C and O when those electrons are explicitly present; this choice is consistent with most previously reported RPA calculations on bulk Cu and CO/Cu(111). For both E^{HF} and $\Delta E^{\text{ACFD-X}}$, the integrable divergence in the electron-repulsion integrals is handled using the Ewald probe-charge approach,[6] which amounts to shifting the total energy by $\frac{n_e v_M}{2}$, where n_e is the total number of electrons per cell and v_M is the Madelung constant.[7] Density fitting is used for both molecular and periodic calculations to approximate the electron repulsion integrals.[8,9] Auxiliary basis sets are optimized for the MCAO-cc-pVXZ orbital basis sets as described in section IV.

B. Quantum Espresso

Quantum Espresso (QE) is used to perform the plane-wave (PW) reference calculations for *fcc* Cu with the four pseudopotentials/ECPs summarized in table S1. For the GTH-PBE-lc pseudopotential and the two ccECPs, the Unified Pseudopotential Format (UPF) files are taken directly from QE's pseudopotential library (for GTH-PBE-lc) or from the QMCPACK `pseudopotentiallibrary` GitHub repository (for the two ccECPs). For the GTH-PBE-sc pseudopotential, we generate the UPF file directly from the pseudopotential parameters using the open-source `gth2upf` library developed by one of the authors.[10] We verified for GTH-PBE-lc that the standard UPF file and the UPF file generated by `gth2upf` give identical results. For each pseudopotential/ECP, the kinetic-energy cutoff is chosen sufficiently large to converge the lattice constant and bulk modulus to better than 0.001 Å and 0.1 GPa, respectively, which is adequate for benchmarking the Gaussian-basis results.

TABLE S1. Pseudopotentials and ECPs used for the QE calculations.

Pseudopotential/ECP	UPF file	Link	`ecutwfc`
GTH-PBE-lc	Cu.pbe-d-hgh.UPF	link	300 Ry
GTH-PBE-sc	Converted by `gth2upf`	link	600 Ry
ccECP-soft	Cu.ccECP-soft.upf	link	300 Ry
ccECP-hard	Cu.ccECP.upf_deprecated	link	1000 Ry

C. VASP

VASP PBE calculations are used for bulk Cu and CO/Cu(111) to assess convergence to the thermodynamic limit (TDL). The single-point calculations use the `Cu_sv_GW`, `C_h`, and `O_h` POTCAR files for Cu, C, and O, respectively, with `ENCUT` set to 1000 eV. For CO/Cu(111), VASP is also used to optimize the geometry at the PBE level. These geometry optimizations use the default `Cu`, `C`, and `O` POTCAR files with `ENCUT` set to 500 eV.

III. MCAO-CC-PVXZ BASIS SETS

The construction of the MCAO-cc-pVXZ basis sets for a given valence-set structure is outlined in sections III A and III B, while the selection of the valence-set structure is discussed in section III D.

A. Atomic optimization

We first perform atomic optimization to generate standard cc-pVXZ basis sets, which serve as the initial guess for the subsequent MCAO. For Cu, this proceeds in three steps by minimizing either the restricted open-shell HF (ROHF) energy or the unrestricted MP2 correlation energy evaluated from the ROHF reference. Both energies are evaluated for the [Ar]$3d^{10}4s^1$ ground state of Cu, and the correlation energy is calculated with the [Ar] core kept frozen.

1. The atomic ROHF energy is minimized to determine the exponents of primitives that are occupied in the ROHF ground state.

2. The MP2 correlation energy is minimized to determine the exponents of primitives for the valence $4p$ shell, which is unoccupied in the ROHF ground state. The exponents optimized from step 1 are kept fixed during this step.

3. The MP2 correlation energy is minimized to determine the exponents of the polarization set, with the valence set exponents from the first two steps kept fixed.

B. MCAO

Starting from a given atomically optimized basis set, MCAO follows the same three-step protocol, but each step includes a condition-number penalty evaluated with the full basis set at the current exponents. For Cu in particular, we have

$$\begin{aligned}
\boldsymbol{\alpha}_{\mathrm{val}}^{(n+1)} &\leftarrow \min_{\boldsymbol{\alpha}_{\mathrm{val}}} E_{\mathrm{atom}}^{\mathrm{HF}}(\boldsymbol{\alpha}_{\mathrm{val}}) + \frac{\varepsilon_{0,\mathrm{val}}}{\kappa_0} \times \max_{M \in \{M\}} \kappa(\boldsymbol{\alpha}_{\mathrm{val}}, \boldsymbol{\alpha}_{4p}^{(n)}, \boldsymbol{\alpha}_{\mathrm{pol}}^{(n)}; M) \\
\boldsymbol{\alpha}_{4p}^{(n+1)} &\leftarrow \min_{\boldsymbol{\alpha}_{4p}} E_{\mathrm{atom}}^{\mathrm{MP2,corr}}(\boldsymbol{\alpha}_{\mathrm{val}}^{(n+1)}, \boldsymbol{\alpha}_{4p}) + \frac{\varepsilon_{0,4p}}{\kappa_0} \times \max_{M \in \{M\}} \kappa(\boldsymbol{\alpha}_{\mathrm{val}}^{(n+1)}, \boldsymbol{\alpha}_{4p}, \boldsymbol{\alpha}_{\mathrm{pol}}^{(n)}; M) \\
\boldsymbol{\alpha}_{\mathrm{pol}}^{(n+1)} &\leftarrow \min_{\boldsymbol{\alpha}_{\mathrm{pol}}} E_{\mathrm{atom}}^{\mathrm{MP2,corr}}(\boldsymbol{\alpha}_{\mathrm{val}}^{(n+1)}, \boldsymbol{\alpha}_{4p}^{(n+1)}, \boldsymbol{\alpha}_{\mathrm{pol}}) + \frac{\varepsilon_{0,\mathrm{pol}}}{\kappa_0} \times \max_{M \in \{M\}} \kappa(\boldsymbol{\alpha}_{\mathrm{val}}^{(n+1)}, \boldsymbol{\alpha}_{4p}^{(n+1)}, \boldsymbol{\alpha}_{\mathrm{pol}}; M)
\end{aligned} \tag{2}$$

Because the κ penalty couples all three exponent subsets, the three optimizations in eq. (2) are iterated to convergence, with $\boldsymbol{\alpha}^{(n)}$ denoting the exponents at the nth macrocycle. We use three macrocycles for all MCAO optimizations in this work, which are sufficient to converge the atomic energy to reasonable precision. The ε_0 parameter is set to 1, 0.1, and 0.01 mE_h for the valence, $4p$, and polarization optimizations, respectively.

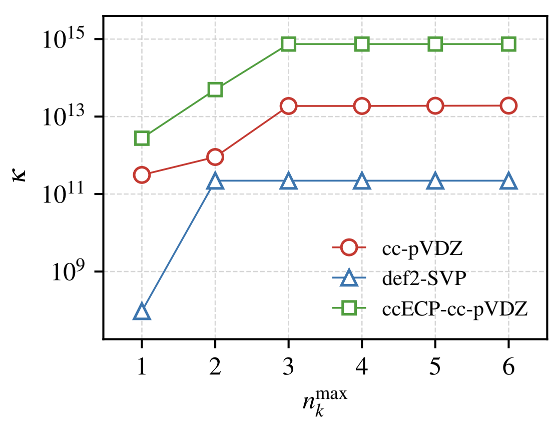

FIG. S1. Convergence of the condition number of various basis sets with respect to the k-mesh size for *fcc* Cu.

For a given reference material, the condition number κ is defined as

$$\kappa_{\mathcal{K}} = \frac{\max_{\boldsymbol{k} \in \mathcal{K}} \max_i |\lambda_{\boldsymbol{k}i}|}{\min_{\boldsymbol{k} \in \mathcal{K}} \min_i |\lambda_{\boldsymbol{k}i}|} \tag{3}$$

where $\lambda_{\boldsymbol{k}i}$ are the eigenvalues of the basis-set overlap matrix evaluated at k-point $\boldsymbol{k}$ for the reference solid. Thus, κ depends on the k-point set $\mathcal{K}$ used to sample the first Brillouin zone. In our implementation, we first determine the maximum k-mesh dimensions along the three reciprocal-lattice directions, $(N_1^{\max}, N_2^{\max}, N_3^{\max})$, such that the corresponding $N_1^{\max} \times N_2^{\max} \times N_3^{\max}$ Born–von Karman supercell is close to cubic and contains more than 300 atoms, i.e.,

$$N_1^{\max} N_2^{\max} N_3^{\max} n_{\text{atom}} \geq 300 \tag{4}$$

where n_{atom} is the number of atoms per unit cell. For *fcc* Cu, $n_{\text{atom}} = 4$ and we set $N_1^{\max} = N_2^{\max} = N_3^{\max} = n_k^{\max}$ for its cubic symmetry. We then determine $\mathcal{K}$ by collecting all symmetrically unique k-points from all possible Γ-centered k-meshes of size $N_1 \times N_2 \times N_3$, where $1 \leq N_\alpha \leq N_\alpha^{\max}$ for $\alpha = 1,2,3$. Figure S1 shows that for *fcc* Cu, κ converges very rapidly and $n_k^{\max} = 3$ essentially saturates κ. Throughout this work, we use a relatively conservative value of $n_k^{\max} = 5$.

To avoid numerical problems caused by an excessively large κ penalty, we adjust κ_0 dynamically during MCAO. In particular, we select initial basis sets whose condition numbers are within 10^2 of the final value of κ_0. In this work, the final value of κ_0 is 10^9, so the initial basis sets have condition numbers below 10^{11}. MCAO then starts with $\kappa_0 = 10^{11}$, so that the κ penalty is at most ε_0 in magnitude. We progressively decrease κ_0 by half a logarithmic unit, i.e., $10^{10.5}, 10^{10}, \cdots$, until reaching the final target $\kappa_0 = 10^9$. At each κ_0, we perform three macro iterations of eq. (2) as discussed above and use the optimized exponents as input for the next κ_0 value. This continuation procedure gives a relatively smooth decrease of κ for the optimized basis sets, as shown in the upper row of fig. S2. Some nonsmooth behavior can occur in the early stages, when κ is still high, likely because the optimization is more susceptible to local minima. The reduction in κ is driven by a continuous increase in the exponents of the most diffuse primitive functions, as shown in the bottom row of fig. S2.

C. Contraction of valence primitives

After all exponents are optimized, the valence set (i.e., the exponents optimized in steps 1 and 2 in section III A) is contracted using the coefficients of atomic natural orbitals (ANOs) calculated at the ROHF level,

$$\phi_{l\mu m}(\boldsymbol{r}) = \sum_{i=1}^{N_l} \chi_{lim}(\boldsymbol{r}) C_{li\mu} \tag{5}$$

where $\mathbf{C}_l$ is an $N_l \times n_l$ coefficient matrix that contracts N_l primitive basis functions $\{\chi_{lim}\}$ within the angular-momentum channel l into n_l contracted basis functions $\{\phi_{l\mu m}\}$ in the same channel.

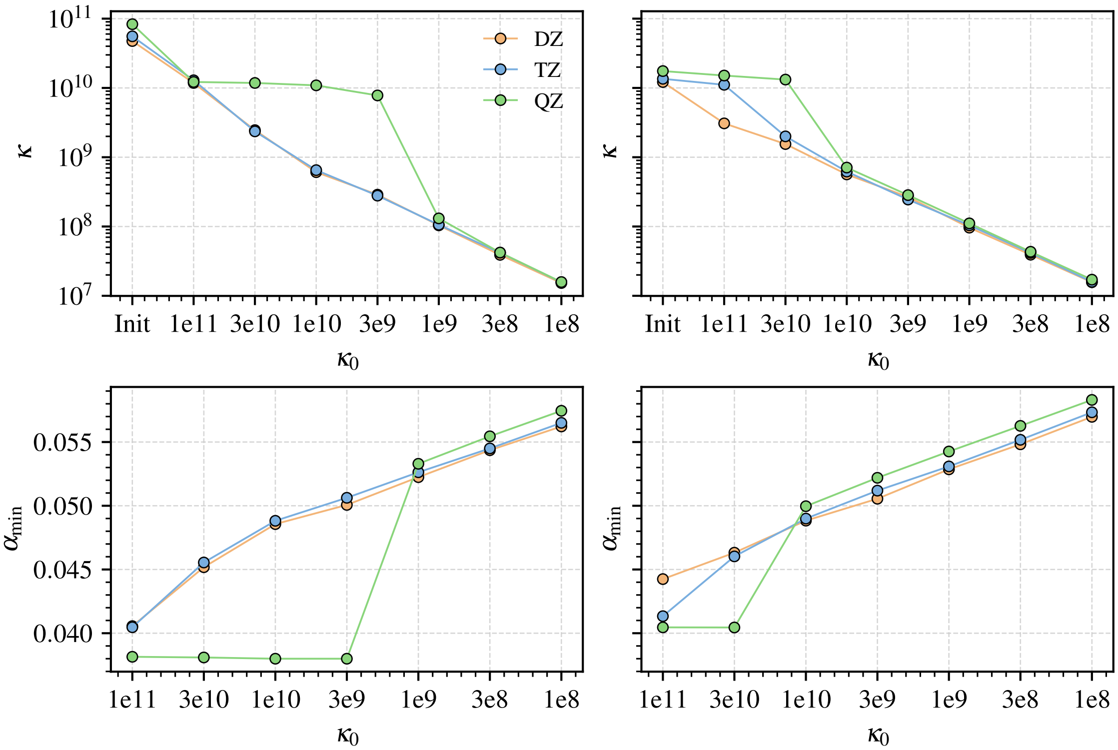

FIG. S2. Variation of the condition number (upper row) and minimum primitive exponent (bottom row) as a function of the applied κ_0 parameter during MCAO. The left two panels are results for the nonrelativistic all-electron basis sets, while the right two panels are results for the SFX2C-1e all-electron basis sets. In all cases, the minimum exponent comes from the s-channel.

The contraction coefficients are determined by solving the generalized eigenvalue equation

$$\mathbf{s}_l \mathbf{d}_l \mathbf{s}_l \mathbf{C}_l = \mathbf{s}_l \mathbf{C}_l \mathbf{\Lambda}_l \tag{6}$$

and retaining eigenvectors $C_{li\mu}$ with nonzero eigenvalues $\lambda_{l\mu}$. In eq. (6), $\mathbf{s}_l$ and $\mathbf{d}_l$ are the spherically averaged radial parts of the atomic overlap and density matrices, respectively,

$$\begin{aligned} s_{lij} &= \frac{1}{2l+1} \sum_{m=-l}^{l} S_{lim,ljm}, \\ d_{lij} &= \frac{1}{2l+1} \sum_{m=-l}^{l} D_{lim,ljm}, \end{aligned} \tag{7}$$

This ANO approach naturally gives the number of contracted GTOs in each shell for the chosen nuclear potential. For example, four, two, and one s, p, and d-type orbitals are generated automatically for all-electron basis sets, while the numbers become two, one, and one for small-core pseudopotentials/ECPs with 19 valence electrons (i.e., $3s^2 3p^6 3d^{10} 4s^1$). The most diffuse one, two, and three primitives in the s, p, and d shells are then decontracted from the fully contracted valence sets to make the final DZ, TZ, and QZ valence sets. For the p shell, the primitive optimized for the unoccupied $4p$ shell is therefore retained outside the ROHF ANO contraction. For the polarization set, we use $1f$, $2f1g$, and $3f2g1h$ for DZ, TZ, and QZ, respectively, which is a natural transition-metal analogue of the $1d$, $2d1f$, and $3d2f1g$ hierarchy for main-group elements. As discussed in the main text, our TZ and QZ polarization sets have the same structure as the molecular cc-pVTZ and cc-pVQZ basis sets, while our DZ is similar to that of def2-SVP.

D. Valence set size selection

The valence-set sizes used in this work for different nuclear potentials and zeta levels are summarized in table S2. For all nuclear potentials, the TZ and QZ basis sets share the same valence primitive set (but contracted differently), which we found to improve extrapolation to the CBS limit. For the DZ valence set, using the same valence primitive set as in TZ and QZ leads to larger RPA@PBE errors for bulk Cu. We therefore vary the size of the d shell and choose the DZ valence set that gives the smallest RPA@PBE error in bulk Cu benchmarks relative to the extrapolated CBS(TZ,QZ) reference. For all nuclear-potential and relativistic treatments considered here, this selection gives a DZ valence set with a slightly smaller d shell than the corresponding TZ and QZ sets. Table S2 also lists the minimum primitive exponent in each valence angular-momentum channel and the condition number of the full basis set, including the polarization functions, evaluated for *fcc* Cu. As discussed in the main text, the selected valence-set sizes maintain similar diffuse primitives within each angular-momentum channel, which in turn gives comparable condition numbers for the reference solid.

IV. AUXILIARY BASIS SETS

For each AO basis set, we optimize an auxiliary basis set for accurate and efficient density-fitting calculations. The auxiliary basis is chosen to be even-tempered and fully uncontracted, so within each angular-momentum channel the Gaussian exponents are generated as

$$\alpha_{l,i} = \alpha_l \beta_l^{i-1}, \qquad i = 1, 2, \cdots, N_l \tag{8}$$

TABLE S2. Valence-set structures (primitive and contracted) used in this work for different nuclear-potential and relativistic treatments. The minimum valence exponents from s, p, and d channels and the condition number of the final MCAO-cc-pVXZ basis sets (including the polarization set) evaluated on *fcc* Cu are also shown for comparison.

	Zeta	Primitive	Contracted	N_{AO}	$\alpha^s_{\min}$	$\alpha^p_{\min}$	$\alpha^d_{\min}$	κ
	DZ	$3s(2+1)p3d$	$2s2p2d$	25	0.048	0.224	0.401	4.57×10^4
GTH-PBE-lc	TZ	$3s(2+1)p4d$	$3s3p3d$	50	0.052	0.207	0.222	8.73×10^7
	QZ	$3s(2+1)p4d$	$3s3p4d$	82	0.052	0.207	0.222	2.29×10^8
	DZ	$5s(4+1)p4d$	$3s2p2d$	26	0.047	0.194	0.308	6.33×10^5
GTH-PBE-sc	TZ	$5s(4+1)p5d$	$4s3p3d$	51	0.051	0.187	0.200	1.65×10^8
	QZ	$5s(4+1)p5d$	$5s4p4d$	87	0.051	0.187	0.200	6.74×10^8
	DZ	$6s(4+1)p4d$	$3s2p2d$	26	0.047	0.201	0.334	1.72×10^6
ccECP-soft	TZ	$6s(4+1)p5d$	$4s3p3d$	51	0.050	0.192	0.219	1.91×10^8
	QZ	$6s(4+1)p5d$	$5s4p4d$	87	0.050	0.192	0.219	6.21×10^8
	DZ	$6s(5+1)p5d$	$3s2p2d$	26	0.046	0.196	0.344	1.40×10^6
ccECP-hard	TZ	$6s(5+1)p6d$	$4s3p3d$	51	0.049	0.187	0.239	1.74×10^8
	QZ	$6s(5+1)p6d$	$5s4p4d$	87	0.049	0.187	0.239	6.16×10^8
	DZ	$16s(10+1)p5d$	$5s3p2d$	31	0.048	0.197	0.389	1.38×10^6
All-e/NR	TZ	$16s(10+1)p7d$	$6s4p3d$	56	0.053	0.184	0.221	8.46×10^7
	QZ	$16s(10+1)p7d$	$7s5p4d$	92	0.053	0.185	0.225	1.23×10^8
	DZ	$17s(10+1)p5d$	$5s3p2d$	31	0.049	0.207	0.385	1.79×10^6
All-e/SFX2C-1e	TZ	$17s(10+1)p7d$	$6s4p3d$	56	0.053	0.193	0.219	8.44×10^7
	QZ	$17s(10+1)p7d$	$7s5p4d$	92	0.054	0.193	0.218	1.04×10^8

The parameters $\boldsymbol{p} = \{\alpha_l, \beta_l\}_{l=0}^{l_{\max}}$ are determined by minimizing a loss function that combines HF and MP2 fitting errors,

$$\mathcal{L}_{\text{aux}}(\boldsymbol{p}) = \max\{\gamma\Delta\mathbf{J}^{\text{HF}}, \gamma\Delta\mathbf{K}^{\text{HF}}, \Delta E_J^{\text{HF}}, \Delta E_K^{\text{HF}}, \Delta E_{\text{corr}}^{\text{MP2}}, \Delta\mathbf{T}_2^{\text{MP2}}\}. \tag{9}$$

Here,

- $\Delta\mathbf{J}^{\text{HF}}$ and $\Delta\mathbf{K}^{\text{HF}}$ are the maximum absolute errors of the HF Coulomb and exchange matrices,
- ΔE_J and ΔE_K are the absolute errors of the HF Coulomb and exchange energies,
- $\Delta E_{\text{corr}}^{\text{MP2}}$ is the absolute error of the MP2 correlation energy,
- and $\Delta\mathbf{T}_2^{\text{MP2}}$ is the maximum absolute error of the MP2 T_2 amplitudes.

All terms in eq. (9) are in atomic units and evaluated on a single atom (for Cu, we use the [Ar]$3d^{10}4s^1$ configuration). Because commonly used auxiliary basis sets often have relatively large $\Delta\mathbf{J}^{\text{HF}}$ and $\Delta\mathbf{K}^{\text{HF}}$, we set $\gamma = 0.1$ to place a looser tolerance on these two matrix-error terms.

Minimizing the auxiliary loss in eq. (9) determines the optimal parameters for an auxiliary basis set of fixed size. For practical density-fitting calculations, the auxiliary basis should be as compact as possible while maintaining the required fitting precision. We therefore start from a sufficiently large set whose optimized loss is below $\tau_{\text{aux}} = 10^{-5}$ and iteratively prune the set until the optimized loss exceeds τ_{aux}. At each iteration, we loop over each angular momentum channel, discard one primitive and reoptimize the exponents of the remaining primitives in that channel while keeping the parameters of all other channels fixed. This produces a pool of optimized candidate sets, each with one fewer primitive than the original set in a single angular-momentum channel. The candidate with the lowest optimized loss is accepted, and the parameters of the full set are reoptimized before the next pruning iteration.

Figure S3 shows the loss of the auxiliary basis sets optimized for the MCAO-cc-pVXZ basis sets with different pseudopotentials, ECPs, and all-electron potentials. The optimized auxiliary bases achieve the target loss of 10^{-5} a.u. for all nuclear-potential treatments and AO ζ levels. For comparison, we also include two commonly used automatic choices: an even-tempered basis (ETB) with $\beta = 2$ and the AutoAux basis developed by Neese and co-workers,[11] both generated using their PySCF implementations. ETB is the default auxiliary basis in PySCF when a named auxiliary basis is unavailable for a given element, while AutoAux is the default in ORCA.[12]

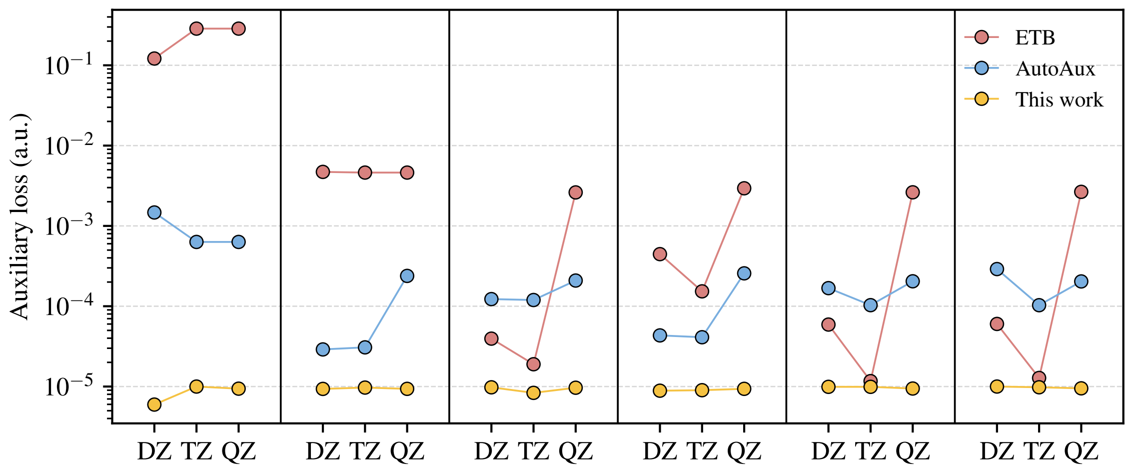

FIG. S3. Comparison of the auxiliary loss defined in eq. (9) calculated using different auxiliary basis sets for a single Cu atom. From left to right, results are shown for different nuclear-potential and relativistic treatments, including GTH-PBE-lc, GTH-PBE-sc, ccECP-hard, ccECP-soft, all-e/NR, and all-e/SFX2C-1e.

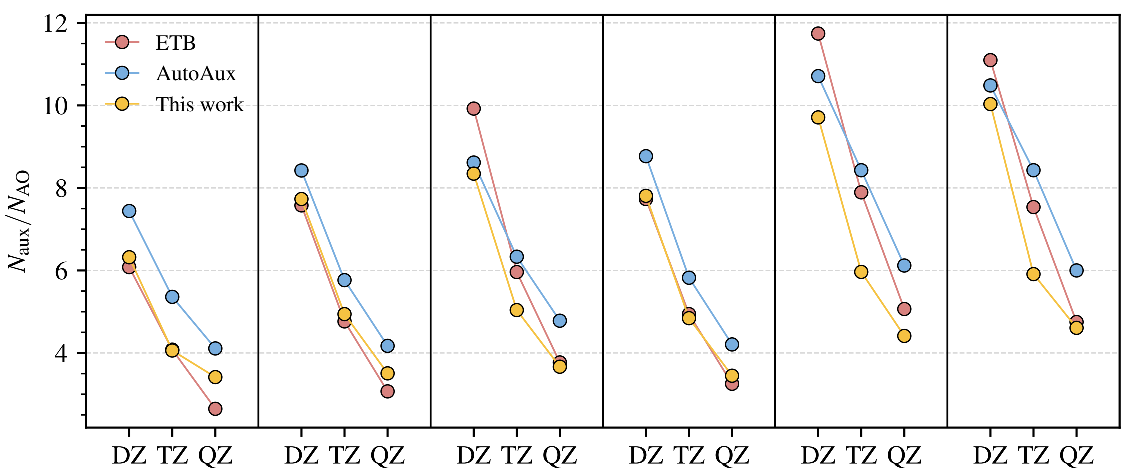

FIG. S4. Comparison of auxiliary-basis-set sizes for a single Cu atom, measured as ratios relative to the corresponding AO basis sets. From left to right, results are shown for different nuclear-potential and relativistic treatments, including GTH-PBE-lc, GTH-PBE-sc, ccECP-hard, ccECP-soft, all-e/NR, and all-e/SFX2C-1e.

As shown in fig. S3, these automatic auxiliary bases either have substantially larger errors than 10^{-5} a.u. or show fluctuating precision across ζ levels and nuclear potentials. The optimized auxiliary bases used in this work are also more compact than the automatic alternatives in almost all cases, as shown in fig. S4.

V. *FCC* BULK COPPER

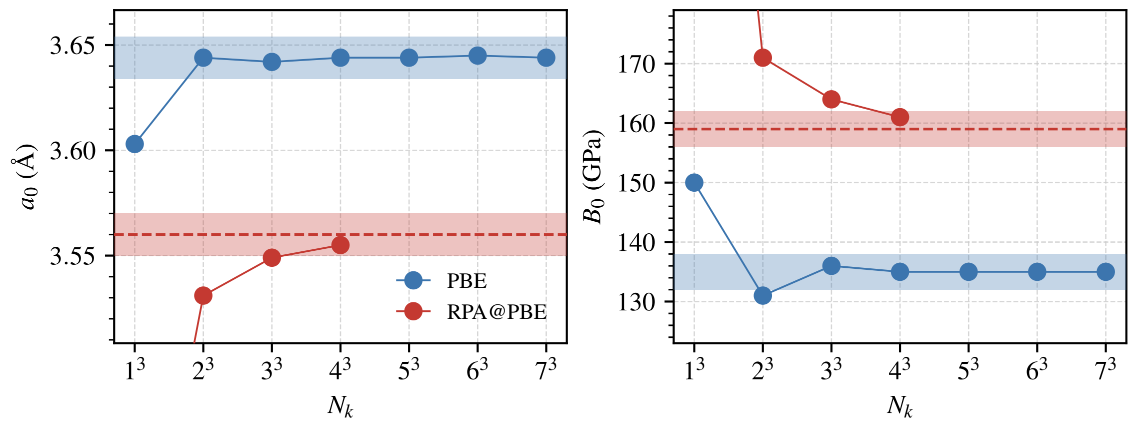

FIG. S5. Convergence of the PBE and RPA@PBE lattice constant (a_0) and bulk modulus (B_0) calculated for *fcc* Cu using the MCAO-cc-pVDZ basis sets optimized for the GTH-PBE-sc pseudopotential. For PBE, the blue shaded area indicates deviation from $N_k = 7^3$ within 0.01 Å for a_0 and 3 GPa for B_0. For RPA@PBE, the red shaded area indicates deviation from the TDL extrapolated results within 0.01 Å for a_0 and 3 GPa for B_0.

The lattice constant a_0 and bulk modulus B_0 are obtained by fitting single-point energies calculated at lattice constants near the experimental value to the Birch–Murnaghan equation of state. For the conventional *fcc* cell (four Cu atoms per cell) used in this work, the Brillouin zone is sampled by $n \times n \times n$ evenly spaced k-point meshes of size $N_k = n^3$. These k-meshes are shifted by the Baldereschi mean-value point[13] (MVP)

$$\boldsymbol{k}_{\mathrm{MVP}}^{fcc} = (0.6223, 0.2953, 0) \tag{10}$$

to reduce the finite-size errors (FSEs). Figure S5 shows the convergence of both a_0 and B_0 with N_k for PBE and RPA@PBE, evaluated using the MCAO-cc-pVDZ basis set with the GTH-PBE-sc pseudopotential. For PBE, $N_k = 4^3$ converges a_0 within 0.01 Å and B_0 within 1 GPa relative to the $N_k = 7^3$ reference. We therefore use $N_k = 4^3$ without further extrapolation. RPA@PBE converges more slowly than PBE with respect to N_k. We obtain the TDL estimates of a_0 and B_0 by first extrapolating the total RPA@PBE energies from $N_k = 3^3$ and $N_k = 4^3$ to the TDL using the $O(N_k^{-1})$ FSE scaling[14] and then fitting the resulting energy curve to the Birch–Murnaghan equation of state. In fig. S5, the extrapolated RPA@PBE a_0 and B_0 are shown as horizontal dashed lines. The largest finite-k-mesh results from $N_k = 4^3$ are within 0.01 Å and 3 GPa of the TDL-extrapolated values, supporting the reliability of the extrapolation.

After obtaining TDL energies in the DZ basis set, we apply the following basis-set composite correction to reach the combined CBS and thermodynamic limit for both PBE and RPA@PBE:

$$E(\mathrm{CBS},\mathrm{TDL}) = E(\mathrm{DZ},\mathrm{TDL}) + E(\mathrm{CBS(T,Q)},N_k^*) - E(\mathrm{DZ},N_k^*) \tag{11}$$

where $N_k^* = 2^3$ with the same Baldereschi MVP shift. For the PBE energy and the mean-field part of the RPA energy, we extrapolate to the CBS limit using the exponential formula from ref 15–18:

$$E(\mathrm{CBS}) = \frac{\exp(\beta X_1)E(X_1) - \exp(\beta X_2)E(X_2)}{\exp(\beta X_1) - \exp(\beta X_2)} \tag{12}$$

where $\beta = 1.54$ as suggested by Karton and Martin,[17] while for the RPA correlation energy, we use the standard Helgaker X^{-3} formula:[19]

$$E(\mathrm{CBS}) = \frac{X_1^3 E(X_1) - X_2^3 E(X_2)}{X_1^3 - X_2^3} \tag{13}$$

In both cases, $X_1 = 3$ and $X_2 = 4$ for TZ and QZ, respectively. The final CBS and TDL values of a_0 and B_0 are listed in table S3.

TABLE S3. Final a_0 (in Å) and B_0 (in GPa) calculated using different nuclear-potential and relativistic treatments, extrapolated to the CBS limit using the MCAO-cc-pV(T,Q)Z basis sets.

Method	PBE		RPA@PBE	
	a_0	B_0	a_0	B_0
GTH-PBE-lc	3.647	134	3.508	218
GTH-PBE-sc	3.629	141	3.592	152
ccECP-soft	3.631	137	3.588	148
ccECP-hard	3.649	133	3.603	145
All-e/NR	3.663	127	3.632	132
All-e/SFX2C-1e	3.623	142	3.591	147

For the TZV2P-MOLOPT and def2-SVP results shown in Fig. 4 of the main text, only the TDL extrapolation is applied as described above; no basis-set extrapolation is performed.

For the PBE band structure, we first perform an SCF calculation using a Baldereschi-MVP-shifted $3 \times 3 \times 3$ k-point mesh and then calculate the bands along the chosen path using non-self-consistent calculations.

VI. CO ADSORPTION ON CU(111)

A. Final E_{ads}

Table S4 lists the final adsorption energies for the top and hollow sites, converged to the combined basis-set and thermodynamic limit using the protocol detailed below. The surface model and geometry protocol largely follow ref 1; the main differences in the present work are the different approach used to converge to the TDL and the use of the MCAO basis sets developed here.

TABLE S4. Final adsorption energies (eV) of CO/Cu(111). The RPA@PBE values are the numbers reported in Fig. 5 of the main text.

Method	PBE			RPA@PBE		
	Top	Hollow	Difference	Top	Hollow	Difference
GTH-PBE-lc	−0.73	−0.86	−0.12	−0.77	−0.75	0.01
GTH-PBE-sc	−0.77	−0.91	−0.14	−0.37	−0.25	0.12
ccECP-soft	−0.76	−0.89	−0.13	−0.35	−0.24	0.11
ccECP-hard	−0.71	−0.83	−0.12	−0.30	−0.18	0.12
All-e/NR	−0.72	−0.85	−0.12	−0.25	−0.13	0.12
All-e/SFX2C-1e	−0.79	−0.94	−0.14	−0.34	−0.24	0.10
PAW/PW (this work)	−0.78	−0.91	−0.13			

B. Geometry

Following previous work,[1] the Cu(111) surface is simulated using a slab model with a 2×2 supercell in the xy plane, four atomic layers, and 15 Å of vacuum in the z direction. The surface is first created using the bulk PBE lattice constant $a_0 = 3.634$ Å. A single CO molecule is then added to either the top site or the *fcc* hollow site, followed by geometry relaxation using a $6 \times 6 \times 1$ unshifted k-point mesh in VASP. The bottom two layers are fixed during the geometry relaxation, while all other atoms are fully relaxed. Dispersion corrections are not used in either the geometry relaxations or the subsequent PBE single-point energy calculations, in order to remain consistent with previous work.[20,21]

C. Adsorption energy

Following ref,[1,22] the adsorption energy is calculated as

$$E_{\text{ads}} = E_{\text{surf+mol}} - E_{\text{surf}} - E_{\text{mol}} \approx E_{\text{int}} + \Delta_{\text{geom}} \tag{14}$$

where $E_{\text{surf+mol}}$, E_{surf}, and E_{mol} are energies of the adsorption system, the free surface, and the free molecule, respectively. The adsorption energy is separated into two parts: E_{int} is the adiabatic interaction energy calculated by freezing the surface and molecule at their adsorption geometry, and Δ_{geom} is the energy change associated with geometry relaxation of the isolated fragments. This decomposition has two practical advantages.

- First, basis-set superposition error (BSSE) can be corrected using the standard counterpoise (CP) approach[23] in the interaction-energy calculation,

  $$E_{\text{int}} = E_{\text{surf+mol}} - E_{\text{surf@surf+mol}} - E_{\text{mol@surf+mol}} \tag{15}$$

  where $E_{\text{surf@surf+mol}}$ and $E_{\text{mol@surf+mol}}$ are the energies of the surface and molecule, respectively, evaluated at their frozen adsorption geometries using the combined surface+molecule basis set.

- Second, previous work shows that surface relaxation is negligible,[1,24] so Δ_{geom} includes only relaxation of the CO molecule and is evaluated at the PBE level, i.e.,

  $$\Delta_{\text{geom}} = E^{\text{PBE}}_{\text{CO(ads)}} - E^{\text{PBE}}_{\text{CO(g)}} \tag{16}$$

  where CO(ads) and CO(g) denote CO in the adsorption and gas-phase geometries, respectively. The CO geometry is characterized by its bond length. We use 1.157 Å, 1.181 Å, and 1.135 Å for the top-site, hollow-site, and gas-phase geometries, respectively, which are in very good agreement with the values used in ref 1. The final relaxation energies extrapolated to the CBS limit using TZ and QZ basis sets for different nuclear-potential and relativistic treatments are listed in table S5.

TABLE S5. CO relaxation energy (eV) defined in eq. (16), evaluated at PBE level and extrapolated to the CBS limit using TZ and QZ basis sets.

Method	Top	Hollow
GTH-PBE	0.022	0.099
ccECP	0.034	0.123
All-e/NR	0.023	0.102
All-e/SFX2C-1e	0.023	0.102

D. Converging E_{int} to the combined basis-set and thermodynamic limit

To establish an efficient protocol for computing E_{int} with controlled basis-set and finite-size errors, we calculated E_{int} with several combinations of basis sets and k-point meshes, as summarized in fig. S6, for both PBE and RPA@PBE using the GTH-PBE-sc pseudopotential. As in the bulk calculations, shifting the k-point meshes by an MVP improves the convergence of E_{int} with k-mesh size. We tested two MVPs. The first is the exact Baldereschi MVP for a two-dimensional hexagonal lattice,[13]

$$\boldsymbol{k}_{\text{MVP}}^{hex} = (0.7614, 0.3807, 0) \tag{17}$$

and the second is an approximate Baldereschi-style MVP derived by requiring only the first-shell Fourier component to vanish for a two-dimensional hexagonal lattice,

$$\boldsymbol{k}_{\text{MVP}}^{hex(1)} = (0.6667, 0.6667, 0). \tag{18}$$

Both MVPs converge faster than Γ-centered meshes; we use eq. (18) for the final calculations because it gives slightly faster convergence for the slab model we used.

For PBE, we also performed VASP calculations, which allow us to study the FSE decay using larger k-meshes than are feasible with the Gaussian basis sets. The VASP PBE E_{int} values converge rapidly with k-mesh size, with the finite 3×3 mesh already reaching an accuracy better than 0.05 eV. The Gaussian-basis results obtained with the MCAO-cc-pVXZ sets follow the same convergence pattern as the VASP results, although the maximum k-mesh size accessible in PySCF is more limited. We therefore use the 3×3 k-mesh for the DZ calculations and apply composite corrections at smaller k-meshes to remove the remaining basis-set incompleteness error. Our final

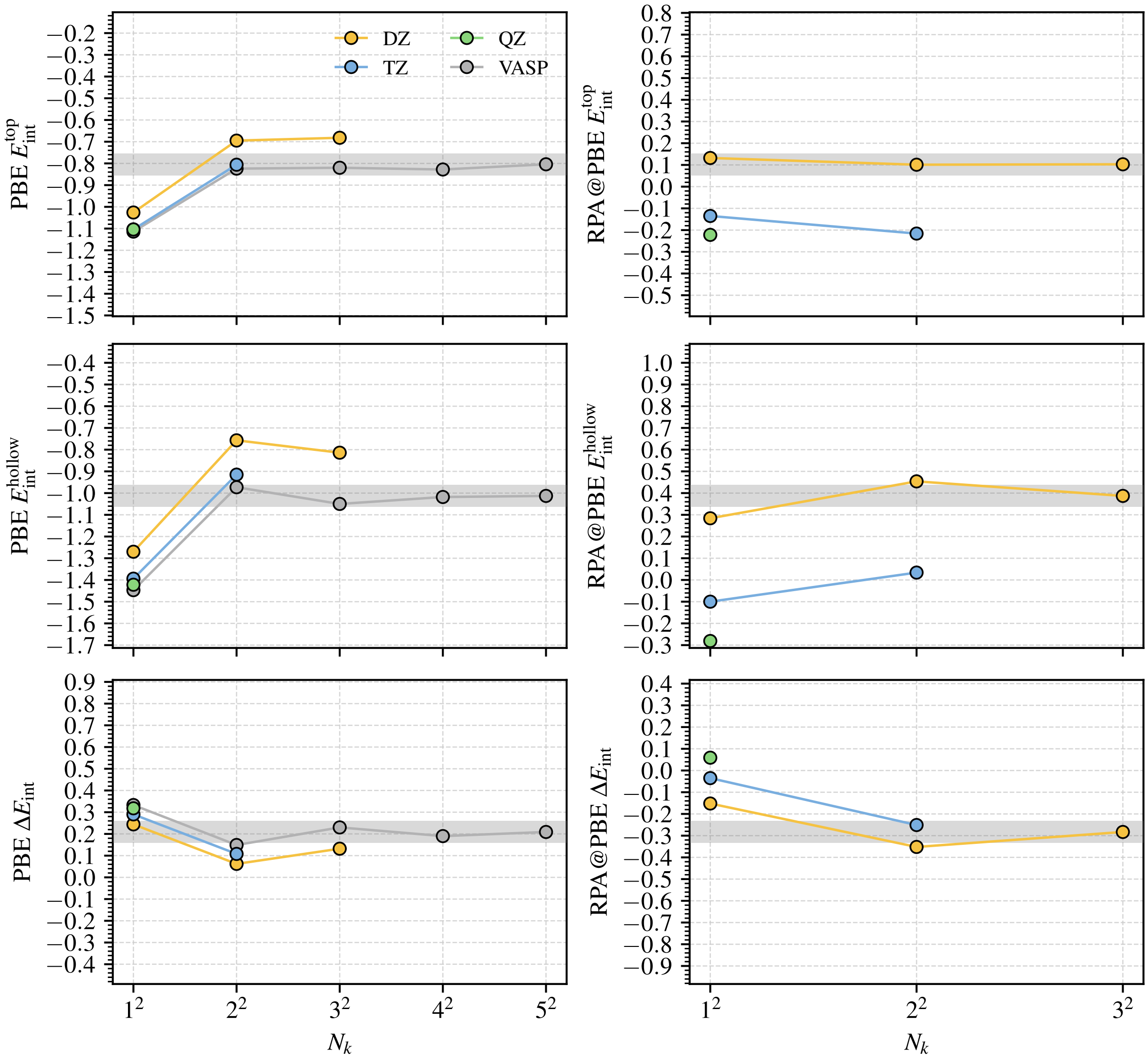

FIG. S6. Convergence of the PBE and RPA@PBE interaction energies with k-mesh size, obtained using the MCAO-cc-pVXZ basis sets with the GTH-PBE-sc pseudopotential. For PBE, VASP PAW/PW results obtained using the structures optimized in this work are also included for comparison. For PBE, the gray shaded area indicates deviations smaller than ± 0.05 eV from the VASP/5×5 results. For RPA@PBE, the gray shaded area indicates deviations smaller than ± 0.05 eV from the DZ/3×3 results, which are the largest-k-mesh RPA calculations performed here.

protocol for PBE E_{int} is as follows:

$$\begin{aligned} E_{\text{int}}(\text{CBS},\text{TDL}) = E_{\text{int}}&(\text{DZ},3\times 3) \\ &+ E_{\text{int}}(\text{TZ},2\times 2) - E_{\text{int}}(\text{DZ},2\times 2) \\ &+ E_{\text{int}}(\text{CBS(T,Q)},1\times 1) - E_{\text{int}}(\text{TZ},1\times 1) \end{aligned} \tag{19}$$

where the CBS(T,Q) extrapolation is performed using eq. (12).

For RPA@PBE, although no PW reference is available, the finite 3×3 mesh also converges the DZ E_{int} to within approximately 0.05 eV, as indicated by the small difference between the 2×2 and 3×3 results. Moreover, the DZ and higher-zeta basis sets show nearly parallel k-mesh convergence, supporting the use of composite corrections at smaller k-meshes for the remaining basis-set incompleteness error. We therefore apply the same protocol in eq. (19) to the RPA@PBE calculations. In this case, the CBS(T,Q) extrapolation is performed separately for the HF@PBE energy and the RPA correlation energy using eqs. (12) and (13), respectively.

E. Results from def2 basis sets

Following ref 1, we also calculated all-electron nonrelativistic E_{ads} using modified def2-SVP and def2-TZVP basis sets in which primitives with exponents below 0.05 a.u. have been removed. To focus on basis-set effects, we use the same PBE-optimized surface models as in the MCAO calculations and approach the combined basis-set and thermodynamic limit using a protocol inspired by eq. (19),

$$\begin{aligned} E_{\text{int}}(\text{CBS},\text{TDL}) = E_{\text{int}}&(\text{DZ},3\times 3) \\ &+ E_{\text{int}}(\text{CBS(D,T)},2\times 2) - E_{\text{int}}(\text{DZ},2\times 2) \end{aligned} \tag{20}$$

where CBS(D,T) is extrapolated using eqs. (12) and (13) with $X_1 = 2$ for def2-SVP and $X_2 = 3$ for def2-TZVP. Using alternative two-point CBS extrapolation formulas does not change the qualitative conclusions below.

The results are listed in table S6. As mentioned in the main text, these values differ from those reported in ref 1 with the same def2-(S,TZ)VP basis sets, likely because finite-size effects are treated differently. Compared with the basis-set-converged all-e/NR results in table S4, the PBE E_{ads} values from the def2-(S,TZ)VP basis sets are reasonable, with errors within 0.05 eV. In contrast, the RPA results have sizable, site-dependent basis-set errors, shifting the predicted site-preference energy by about 0.2 eV. At least part of this error arises from the absence of QZ-quality information as in the CBS(T,Q) extrapolation. Consistent with this interpretation, table S6 also includes results obtained using eq. (20) with our MCAO-cc-pV(D,T)Z basis sets, which are in reasonable agreement with the def2-(S,TZ)VP results.

TABLE S6. All-electron nonrelativistic PBE and RPA@PBE E_{ads} (eV) calculated using eq. (20) for the def2-(S,TZ)VP series and the MCAO-cc-pV(D,T)Z series.

Method	PBE			RPA@PBE		
	Top	Hollow	Difference	Top	Hollow	Difference
def2-(S,TZ)VP	−0.68	−0.78	−0.10	0.08	0.38	0.29
MCAO-cc-pV(D,T)Z	−0.63	−0.71	−0.08	0.13	0.44	0.31

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